

Document 1: The Concept Blueprint (Module 1)

Core Architecture Concepts

- **API (Application Programming Interface):** A secure digital bridge that allows your local application to send requests and receive responses from external services (like Google's AI servers) over the internet.
- **Environment Variables (`.env`):** A secure method for storing sensitive configuration data (like your `GEMINI_API_KEY`) completely separated from your source code. This ensures your private keys are never hardcoded or accidentally shared.
- **JSON (JavaScript Object Notation):** A text-based, standard data-interchange format structured using `{"key": "value"}` pairs. It is the universal language systems use to trade data without the ambiguity of conversational human language.
- **Parsing (`json.loads`):** The process of translating a raw text string formatted as JSON into a native, machine-readable data structure (like a Python Dictionary) so your application can query individual data points.

Technical Stack Used

- `google-generativeai`: The official developer SDK used to initialize communication clients and interface with the `gemini-2.5-flash` model.
- `python-dotenv`: A lightweight library used to scan your project folder for a `.env` file and load its contents into your laptop's environment memory.